

Solutions to Introduction to Topological Manifolds

Chapter 11: Covering Maps

Problem 1 (11-1). Let $q : E \rightarrow X$ be a covering map.

(a) Show that if X is a Hausdorff space, then E is too.

Proof. Let $e_1, e_2 \in E$ be two distinct points ($e_1 \neq e_2$). Denote their projections under the covering map q as $x = q(e_1)$ and $y = q(e_2)$. We split the proof into two exhaustive cases based on whether they belong to the same fiber:

1° **e_1 and e_2 lie in the same fiber (i.e., $x = y$):** Since q is a covering map, there exists an evenly covered open neighborhood $U \subset X$ containing x . By definition, its preimage decomposes into a disjoint union of open subsets in E :

$$q^{-1}(U) = \coprod_{\alpha} V_{\alpha},$$

where each V_{α} is an open set mapped homeomorphically onto U by q .

Since $e_1 \neq e_2$ but they belong to the same fiber $q^{-1}(x)$, they cannot belong to the same local sheet. Thus, there exist distinct indices $\alpha \neq \beta$ such that $e_1 \in V_{\alpha}$ and $e_2 \in V_{\beta}$.

By the definition of a disjoint union, $V_{\alpha} \cap V_{\beta} = \emptyset$. Since V_{α} and V_{β} are open in E , they provide the required disjoint open neighborhoods separating e_1 and e_2 .

2° **e_1 and e_2 lie in different fibers (i.e., $x \neq y$):** Since X is a Hausdorff space and $x \neq y$, there exist disjoint open neighborhoods $U_x, U_y \subset X$ for x and y respectively, satisfying $U_x \cap U_y = \emptyset$.

By the continuity of q , the preimages $q^{-1}(U_x)$ and $q^{-1}(U_y)$ are open subsets in E . Clearly, $e_1 \in q^{-1}(U_x)$ and $e_2 \in q^{-1}(U_y)$.

We claim that these preimages are disjoint. Suppose for contradiction that there exists a point $e_3 \in q^{-1}(U_x) \cap q^{-1}(U_y)$. Applying q yields $q(e_3) \in U_x$ and $q(e_3) \in U_y$, which implies $q(e_3) \in U_x \cap U_y$. This directly contradicts $U_x \cap U_y = \emptyset$. Therefore, $q^{-1}(U_x) \cap q^{-1}(U_y) = \emptyset$.

Since both preimages are open in E , they serve as the desired disjoint open neighborhoods separating e_1 and e_2 .

In either case, e_1 and e_2 can be separated by disjoint open neighborhoods. Consequently, the covering space E is a Hausdorff space. $\square$

(b) Show that if X is an n -manifold, then E is too.

Proof. To show that E is an n -manifold, we must verify that E is a Hausdorff space, locally Euclidean of dimension n , and has a second-countable basis. We check these three conditions sequentially:

1° **Hausdorff:** By part (a), since X is a Hausdorff space and $q : E \rightarrow X$ is a covering map (which is a local homeomorphism), it follows directly that E is also a Hausdorff space.

2° **Locally Euclidean:** Let $e \in E$ be an arbitrary point, and let $x = q(e) \in X$. Since X is an n -manifold, there exists an open neighborhood U' of x that is homeomorphic to an open subset of $\mathbb{R}^n$.

By the definition of a covering map, there exists an evenly covered neighborhood U of x . Let $W = U \cap U'$. Then W is still an open neighborhood of x and is homeomorphic to an open subset of $\mathbb{R}^n$. Since $W \subset U$, its preimage decomposes into a disjoint union of open slices:

$$q^{-1}(W) = \bigsqcup_{\alpha \in A} G_\alpha.$$

There exists a unique slice G_{α_0} containing e . Since $q|_{G_{\alpha_0}} : G_{\alpha_0} \rightarrow W$ is a homeomorphism, and W is homeomorphic to an open subset of $\mathbb{R}^n$, it follows that G_{α_0} is an open neighborhood of e in E that is homeomorphic to an open subset of $\mathbb{R}^n$. Thus, E is locally Euclidean of dimension n .

3° **Second-Countable:** Let $\mathcal{B}$ be a countable basis for the topology of X . Without loss of generality, we can assume that each $K \in \mathcal{B}$ is an evenly covered neighborhood (since evenly covered neighborhoods form a basis for X).

For each $K \in \mathcal{B}$, its preimage decomposes into a disjoint union of open slices:

$$q^{-1}(K) = \bigsqcup_{\alpha \in A_K} (G_K)_\alpha.$$

Since X is second-countable and q is a covering map, the fiber over any point is at most countable, which implies that the index set A_K is at most countable for each $K \in \mathcal{B}$. We define a collection of open sets in E by:

$$\mathcal{B}_E = \{(G_K)_\alpha \mid K \in \mathcal{B}, \alpha \in A_K\}.$$

Since $\mathcal{B}_E$ is a countable union of countable sets, $\mathcal{B}_E$ is a countable collection.

Next, we prove that $\mathcal{B}_E$ is a basis for E . Let $V \subset E$ be an arbitrary open subset and let $e \in V$. Since q is a local homeomorphism, there exists an open neighborhood $V_e \subset V$ of e such that $q|_{V_e} : V_e \rightarrow q(V_e)$ is a homeomorphism. Since $q(e) \in q(V_e)$ and $q(V_e)$ is open in X , there exists a basis element $K \in \mathcal{B}$ such that $q(e) \in K \subset q(V_e)$.

Lifting this inclusion back to the slice containing e , we obtain:

$$e \in (G_K)_{\alpha_0} \subset V_e \subset V,$$

where $(G_K)_{\alpha_0}$ is the unique slice of $q^{-1}(K)$ containing e . Therefore, V can be written as a union of elements from $\mathcal{B}_E$. This proves that $\mathcal{B}_E$ is a countable basis for E , so E is second-countable.

In any case, E satisfies the Hausdorff property, the local Euclidean property of dimension n , and second-countability. Consequently, the covering space E is an n -manifold. $\square$

(c) Show that if E is an n -manifold and X is Hausdorff, then X is an n -manifold.

Proof. To show that X is an n -manifold, we need to verify that X is Hausdorff, locally Euclidean of dimension n , and second-countable.

1° **Hausdorff:** By assumption, X is already a Hausdorff space.

2° Locally Euclidean: Let $x \in X$. Since $q : E \rightarrow X$ is a covering map, there exists an evenly covered neighborhood $U \subset X$ of x . The preimage $q^{-1}(U)$ decomposes into a disjoint union of open slices:

$$q^{-1}(U) = \bigsqcup_{\alpha \in A} G_\alpha,$$

where each slice G_α is homeomorphic to U via the restriction of q .

Since E is an n -manifold, for any point $e \in G_\alpha$, there exists an open neighborhood $V_e \subset G_\alpha$ that is homeomorphic to an open subset of $\mathbb{R}^n$. The homeomorphism from G_α to U allows us to transfer this local Euclidean structure back to U . Thus, every point in U , including x , has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$, proving that X is locally Euclidean of dimension n .

3° Second-Countable: Since E is second-countable, let $\mathcal{B}_E$ be a countable basis for the topology of E . For each basis element $(G_K)_\alpha \in \mathcal{B}_E$, its image under the covering map, $q((G_K)_\alpha)$, is an open set in X . Define:

$$\mathcal{B}_X = \{q((G_K)_\alpha) \mid (G_K)_\alpha \in \mathcal{B}_E\}.$$

Since $\mathcal{B}_E$ is countable and the covering map has at most countable fibers, $\mathcal{B}_X$ is also countable. □

Problem 2 (11-2). Prove that for any $n \geq 1$, the map $q : S^n \rightarrow \mathbb{P}^n$ defined by sending each point $x \in S^n$ to the line through the origin and x is a covering map.

Proof. By definition, the real projective space $\mathbb{P}^n$ is equipped with the quotient topology induced by the map $q : S^n \rightarrow \mathbb{P}^n$, which identifies antipodal points, i.e., $q^{-1}([x]) = \{x, -x\}$ for any $x \in S^n$. Thus, q is naturally continuous and surjective.

To prove that q is a covering map, for any point $[x] \in \mathbb{P}^n$, we choose one of its preimages $x \in S^n \subset \mathbb{R}^{n+1}$. Consider the open upper hemisphere centered at x within S^n , defined as:

$$U = \{y \in S^n \mid y \cdot x > 0\}$$

where $y \cdot x$ denotes the standard inner product in $\mathbb{R}^{n+1}$. Clearly, U is an open neighborhood of x in S^n . Let $-U = \{-y \mid y \in U\}$. By symmetry, $-U$ is the open lower hemisphere centered at $-x$, and it is clear that $U \cap -U = \emptyset$.

Now, let $V = q(U) \subset \mathbb{P}^n$. To show that V is an open neighborhood of $[x]$, we look at its full preimage under q . By the antipodal identification, we have:

$$q^{-1}(V) = q^{-1}(q(U)) = U \cup -U$$

Since both U and $-U$ are open in S^n , their union $q^{-1}(V)$ is also open in S^n . By the definition of the quotient topology, a subset $V \subset \mathbb{P}^n$ is open if and only if $q^{-1}(V)$ is open in S^n . Therefore, V is an open neighborhood of $[x]$ in $\mathbb{P}^n$.

Finally, we examine the restriction $q|_U : U \rightarrow V$.

1° Surjectivity: For any $[y] \in V$, its preimage in $U \cup -U$ must have exactly one representative in U (since U contains exactly one point from each antipodal pair $\{y, -y\}$). Thus, $q|_U$ is bijective.

2° **Homeomorphism:** Since q is a quotient map and $U \cup -U$ is an open saturated set, q is an open map. The restriction of an open continuous map to an open set remains open and continuous. Since $q|_U$ is a bijective, continuous, and open map, it is a homeomorphism from U onto V .

By identical reasoning, $q|_{-U} : -U \rightarrow V$ is also a homeomorphism.

Consequently, for every $[x] \in \mathbb{P}^n$, we have found an open neighborhood V whose preimage $q^{-1}(V) = U \sqcup -U$ is a disjoint union of open sets in S^n , each mapped homeomorphically onto V by q . Thus, $q : S^n \rightarrow \mathbb{P}^n$ is a covering map. □

Problem 3 (11-3). Let S be the following subset of $\mathbb{C}^2$:

$$S = \{(z, w) \in \mathbb{C}^2 \mid w^2 = z, w \neq 0\}$$

Show that the projection $\pi_1 : \mathbb{C}^2 \rightarrow \mathbb{C}$ given by $\pi_1(z, w) = z$ restricts to a two-sheeted covering map $q : S \rightarrow \mathbb{C} \setminus \{0\}$.

Proof. Let $M = \mathbb{C} \setminus \{0\}$. The map $q : S \rightarrow M$ is the restriction of the standard coordinate projection $\pi_1 : \mathbb{C}^2 \rightarrow \mathbb{C}$ to the subspace S .

1° **Continuity:** Since the global projection π_1 is continuous, its restriction $q = \pi_1|_S$ to the subspace S is automatically continuous.

2° **Surjectivity:** For any $z_0 \in M$, by the fundamental theorem of algebra, the equation $w^2 = z_0$ has exactly two distinct non-zero solutions, which we denote as w_0 and $-w_0$. Thus, (z_0, w_0) and $(z_0, -w_0)$ both belong to S , and $q(z_0, \pm w_0) = z_0$. This proves that q is surjective, and each fiber $q^{-1}(z_0)$ consists of exactly two distinct points.

3° **Evenly Covered Neighborhoods:** For any $z_0 \in M$, we can express it in polar coordinates as $z_0 = r_0 e^{i\theta_0}$. We can choose a sufficiently small open disk $U \subset M$ centered at z_0 that does not wrap around the origin (for instance, by restricting the argument to an interval of length less than 2π , say $\theta \in (\theta_0 - \pi, \theta_0 + \pi)$, limit U in single complex plane, so that it does not contain the origin).

On this simply connected domain U , we can define two well-defined, continuous, and holomorphic single-valued branches of the complex square root function, denoted by $g_1 : U \rightarrow \mathbb{C}$ and $g_2 : U \rightarrow \mathbb{C}$, where $g_2(z) = -g_1(z)$. By definition, $(g_1(z))^2 = (g_2(z))^2 = z$ for all $z \in U$.

Now, we define two subsets of S :

$$U_1 = \{(z, g_1(z)) \in \mathbb{C}^2 \mid z \in U\}$$

$$U_2 = \{(z, g_2(z)) \in \mathbb{C}^2 \mid z \in U\}$$

We verify that U_1 and U_2 satisfy the conditions for a covering space:

1. **Openness:** U_1 and U_2 are open in S because they are the graphs of continuous functions g_1, g_2 defined on the open set U . Alternatively, $U_1 = S \cap (U \times W_1)$ where W_1 is an open neighborhood around $g_1(z_0)$ excluding $-g_1(z_0)$.
2. **Disjointness:** Since $g_1(z) \neq g_2(z)$ for any $z \in U$ (as $z \neq 0$), we have $U_1 \cap U_2 = \emptyset$.
3. **Preimage Union:** For any $(z, w) \in q^{-1}(U)$, we have $z \in U$ and $w^2 = z$. Since $g_1(z)$ and $g_2(z)$ are the only two roots of $w^2 = z$, w must equal either $g_1(z)$ or $g_2(z)$. Thus, $(z, w) \in U_1 \cup U_2$, meaning $q^{-1}(U) = U_1 \sqcup U_2$.

4. **Homeomorphism:** The restriction $q|_{U_1} : U_1 \rightarrow U$ maps $(z, g_1(z)) \mapsto z$. Its inverse is given by $z \mapsto (z, g_1(z))$, which is continuous since g_1 is continuous. Thus, $q|_{U_1}$ is a homeomorphism. By identical logic, $q|_{U_2} : U_2 \rightarrow U$ is also a homeomorphism.

Since every point $z_0 \in \mathbb{C} \setminus \{0\}$ has an evenly covered neighborhood U with exactly two disjoint preimages, $q : S \rightarrow \mathbb{C} \setminus \{0\}$ is a two-sheeted covering map. $\square$

Problem 4 (11-4). Show that there is a two-sheeted covering of the Klein bottle by the torus.

Proof. Let $T^2 = S^1 \times S^1$ denote the torus, and let K denote the Klein bottle. Let $q_1 : \mathbb{R}^2 \rightarrow T^2$ be a map satisfying $T^2 = \mathbb{R}^2 / (2\mathbb{Z} \times \mathbb{Z})$, and let $q_2 : \mathbb{R}^2 \rightarrow K$ be a map satisfying $K = \mathbb{R}^2 / \Gamma$, where Γ is the group generated by the transformations $g_1 : (x, y) \mapsto (x+1, -y)$ and $g_2 : (x, y) \mapsto (x, y+1)$. Notice that $g_1^2 : (x, y) \mapsto (x+2, y)$, so the subgroup $2\mathbb{Z} \times \mathbb{Z}$ generated by g_1^2 and g_2 is a normal subgroup of Γ .

Suppose $q : T^2 \rightarrow K$ is the map, since preceding text, we can also define q by group action $q : \mathbb{R}^2 / (2\mathbb{Z} \times \mathbb{Z}) \rightarrow \mathbb{R}^2 / \Gamma$. Now to prove q is a covering map, we need to show that q is well-defined, continuous, surjective, and for every point $k \in K$, there exists an open neighborhood U of k such that $q^{-1}(U)$ is a disjoint union of open sets in T^2 , each mapped homeomorphically onto U by q .

- 1° **Well-definedness:** First, we prove $[z_0]_{\Gamma'} = [z_1]_{\Gamma'} \Leftrightarrow z_1 = \gamma'(z_0)$. As $z \sim z' \iff \exists \gamma' \in \Gamma'$ s.t. $z' = \gamma'(z)$ is an equivalence relation, equivalence class of this is

$$[z_0]_{\Gamma'} := \{\gamma'(z_0) : \gamma' \in \Gamma'\} = \Gamma' \cdot z_0$$

where

$$\Gamma' = \{\gamma_1^m \gamma_2^n : m, n \in \mathbb{Z}\} = \{z \mapsto z + mv_1 + nv_2 : m, n \in \mathbb{Z}\}.$$

Γ' is a group, so it has inverse element $\gamma'^{-1} \in \Gamma'$ such that $\gamma'^{-1}\gamma' = id$. We must show $\Gamma' \cdot z_0 = \Gamma' \cdot z_1$.

1. $[z_1]_{\Gamma'} \subseteq [z_0]_{\Gamma'}$: Take arbitrary element of $[z_1]_{\Gamma'}$, which is $\delta(z_1)$ for some $\delta \in \Gamma'$. Substituting $z_1 = \gamma'(z_0)$:

$$\delta(z_1) = \delta(\gamma'(z_0)) = (\delta\gamma')(z_0)$$

Since Γ' is a group, $\delta\gamma' \in \Gamma'$, hence

$$\delta(z_1) \in \Gamma' \cdot z_0 = [z_0]_{\Gamma'}$$

2. $[z_0]_{\Gamma'} \subseteq [z_1]_{\Gamma'}$: Since Γ' is a group, γ' has an inverse $(\gamma')^{-1} \in \Gamma'$, and

$$z_0 = (\gamma')^{-1}(z_1)$$

(applying $(\gamma')^{-1}$ to both sides of $z_1 = \gamma'(z_0)$). This shows that z_0 can also be obtained from z_1 via an element of Γ' , so the same argument (with the roles of z_0, z_1 swapped) gives

$$[z_0]_{\Gamma'} \subseteq [z_1]_{\Gamma'}$$

So we have

$$z_1 = \gamma'(z_0) \implies [z_0]_{\Gamma'} = [z_1]_{\Gamma'}.$$

Conversely, if $[z_0]_{\Gamma'} = [z_1]_{\Gamma'}$. Since $z_1 \in [z_1]_{\Gamma'}$ (every point lies in its own orbit, taking $\delta = e$, the identity: $e(z_1) = z_1$), and by hypothesis $[z_1]_{\Gamma'} = [z_0]_{\Gamma'}$, we get

$$z_1 \in [z_0]_{\Gamma'}.$$

By the definition of $[z_0]_{\Gamma'}$ above, $z_1 \in [z_0]_{\Gamma'}$ means there exists some $\gamma' \in \Gamma'$ such that

$$z_1 = \gamma'(z_0),$$

which is exactly the desired conclusion. Then we have

$$[z_0]_{\Gamma'} = [z_1]_{\Gamma'} \implies z_1 = \gamma'(z_0).$$

Since $\Gamma' = (2\mathbb{Z} \times \mathbb{Z}) \subset \Gamma$, let $z_0, z_1 \in \mathbb{R}^2$ be arbitrary satisfying $[z_0]_{\Gamma'} = [z_1]_{\Gamma'}$, where exists $\gamma' \in \Gamma' \subset \Gamma$ such that $z_1 = \gamma'(z_0)$ which implies $[z_0]_{\Gamma} = [z_1]_{\Gamma}$. So $q([z_0]_{\Gamma'}) = q([z_1]_{\Gamma'})$, which q is well-defined.

2° **Continuity:** Suppose $V \subset K$ is open. Then we need to prove that $q^{-1}(V)$ is open in T^2 . For every $z \in \mathbb{R}^2$,

$$q(q_1(z)) = q([z]_{\Gamma'}) = [z]_{\Gamma} = q_2(z),$$

so we obtain the following commutative diagram:

$$q \circ q_1 = q_2.$$

Since q_2 is continuous, for any open set $V \subset K$, the preimage $q_2^{-1}(V)$ is open in $\mathbb{R}^2$. We have

$$q_1^{-1}(q^{-1}(V)) = q_2^{-1}(V),$$

since q_1 is a quotient map, $q^{-1}(V) \subset T^2$ is open if and only if $q_1^{-1}(q^{-1}(V))$ is open. So $q^{-1}(V)$ is open in T^2 . Hence, q is continuous.

3° **Surjectivity:** Since q_2 is surjective, for any $[z]_{\Gamma} \in K$, there exists $z \in \mathbb{R}^2$ representing this class (a fixed/based point). Then we can take $[z]_{\Gamma'} \in T^2$, which satisfies $q([z]_{\Gamma'}) = [z]_{\Gamma}$. So q is surjective.

4° **Evenly Covered Neighborhoods:** Since $q \circ q_1 = q_2$ and they are both surjective, for any $[z]_{\Gamma} \in K$, we can get $z \in \mathbb{R}^2$ representing this class and $[z]_{\Gamma'} \in T^2$. Since q_2 is a covering map, there exists an open neighborhood $U \subset K$ of $[z]_{\Gamma}$ such that $q_2^{-1}(U)$ is a disjoint union of open sets in $\mathbb{R}^2$, each mapped homeomorphically onto U by q_2 and the cardinality of any fiber is called the number of sheets of the covering which means that

$$q_2^{-1}(U) = \bigsqcup_{\gamma \in \Gamma} \gamma(V) = \left(\bigsqcup_{\gamma \in \Gamma'} \gamma(V) \right) \sqcup \left(\bigsqcup_{\gamma \in g_1 \Gamma'} \gamma(V) \right) \triangleq W_1 \sqcup W_2.$$

For $z \in \mathbb{R}^2$, there exists a unique open set $V \subset q_2^{-1}(U)$ containing z such that $q_2|_V : V \rightarrow U$ is a homeomorphism. Let $W = q_1(V) \subset T^2$, which is open in T^2 . Since q_1 is a quotient map and $[z]_{\Gamma'} \in W$, similarly, there exists an open neighborhood, without loss of generality assume it is W , such that $q_1^{-1}(W)$ is a disjoint union of open sets in $\mathbb{R}^2$, each mapped homeomorphically onto W by q_1 and

$$q_1^{-1}(W) = \bigsqcup_{\gamma \in \Gamma'} \gamma(V) = W_1.$$

Obviously, $q_1^{-1}(q^{-1}(U)) = q_2^{-1}(U) = W_1 \sqcup W_2$. Let $U_1 = q_1(W_1) = W$ and $U_2 = q_1(W_2) = q_1(g_1(V))$. Now we prove that $U_1 \cap U_2 = \emptyset$. Let w is arbitrary element of $U_1 \cap U_2$, then there exists $w_1 \in W_1$ and $w_2 \in W_2$ such that $q_1(w_1) = w = q_1(w_2)$. Since q_1 is a quotient

map, there exists $\gamma' \in \Gamma'$ such that $w_2 = \gamma'(w_1)$. But W_1 and W_2 are disjoint, so this is impossible. Hence, $U_1 \cap U_2 = \emptyset$. Next to prove that $q|_{U_i} : U_i \rightarrow U$ is a homeomorphism for $i = 1, 2$. Since $q \circ q_1 = q_2$ and $q_1|_{W_1} : W_1 \rightarrow U_1$ is a homeomorphism, $q_2|_{W_1} : W_1 \rightarrow U$ is a homeomorphism too. So $q|_{U_1} = q_2 \circ (q_1|_{W_1})^{-1}$ is a homeomorphism. Similarly, $q|_{U_2} = q_2 \circ (q_1|_{W_2})^{-1}$ is a homeomorphism.

□

Problem 5 (11-10). Show that a covering map is proper if and only if it is finite-sheeted.

Proof. A map which is proper satisfies that the preimage of every compact set is compact. Suppose $q : E \rightarrow X$ is a covering map.

1° **Necessity:** If q is proper, then for any $x \in X$, the singleton $\{x\}$ is compact, so its preimage $q^{-1}(x)$ is compact in E . Since x is evenly covered, there exists an open neighborhood U of x such that $q^{-1}(U) = \bigsqcup_{\alpha \in A} V_\alpha$, where each V_α is open in E and contains unique point $x_\alpha \in q^{-1}(x)$. By the definition of the subspace topology on $q^{-1}(x)$, the intersection

$$\{x_\alpha\} = V_\alpha \cap q^{-1}(x)$$

is open in $q^{-1}(x)$ for each $\alpha \in A$. This implies that $q^{-1}(x)$ is a discrete space. Since $\mathcal{U} = \{\{x_\alpha\} : \alpha \in A\}$ forms an open cover of the compact space $q^{-1}(x)$, it must admit a finite subcover. Thus, the index set A is finite, meaning that q is finite-sheeted.

2° **Sufficiency:** If q is finite-sheeted, then for any compact set $K \subset X$, let $\{\mathcal{W}_i\}$ be an arbitrary open cover of $q^{-1}(K)$. For $x \in K$ is arbitrary, then exists neighborhood U_x of x such that $q^{-1}(U_x) = \bigsqcup_{i=1}^{n_x} G_i^x$, where n_x is a finite index set, each G_i^x is is homeomorphic to U_x via the restriction of q and each $G_i^x \cap q^{-1}(x)$ is a singleton $e_i \in \mathcal{W}_{j(i)}$. So let $\phi_i = (q|_{G_i^x})^{-1} : U_x \rightarrow G_i^x$ is homeomorphism, we define $U_x^i = \phi_i^{-1}(\mathcal{W}_{j(i)} \cap G_i^x) \subset U_x$ is open in U_x and $x \in U_x^i$ since $\phi_i(x) = e_i \in \mathcal{W}_{j(i)}$. Let $U'_x := \bigcap_{i=1}^{n_x} U_x^i$ is open in U_x and $x \in U'_x$ such that each $q^{-1}(U'_x) \cap G_i^x \subset \mathcal{W}_{j(i)}$. So $q^{-1}(U'_x)$ is covered by finite number of open sets $\mathcal{W}_{j(i)}$ for $i = 1, 2, \dots, n_x$ and $\{U'_x : x \in K\}$ is an open cover of K . Since K is compact, there exists a finite subcover $\{U'_{x_1}, U'_{x_2}, \dots, U'_{x_m}\}$ of K , which means $q^{-1}(K) \subseteq \bigcup_{s=1}^m q^{-1}(U'_{x_s})$. Since each $q^{-1}(U'_{x_s}) = \bigsqcup_{i=1}^{n_{x_s}} (q^{-1}(U'_{x_s}) \cap G_i^{x_s}) \subseteq \bigcup_{i=1}^{n_{x_s}} \mathcal{W}_{j_s(i)}$, we have:

$$q^{-1}(K) \subseteq \bigcup_{s=1}^m \bigcup_{i=1}^{n_{x_s}} \mathcal{W}_{j_s(i)}.$$

Since this union consists of only finitely many sets (specifically, $\sum_{s=1}^m n_{x_s}$ sets), we have found a finite subcover of $\{\mathcal{W}_i\}$ for $q^{-1}(K)$. Hence, $q^{-1}(K)$ is compact, and q is proper.

□

Problem 6 (11-11). Let $q : E \rightarrow X$ be a covering map. Show that E is compact if and only if X is compact and q is a finite-sheeted covering.

Proof.

1° **Sufficiency:** If X is compact and q is finite-sheeted, since problem 11-10, q is proper map. The preimage of compact set is compact, so $E = q^{-1}(X)$ is compact.

2° **Necessity:** Since E is compact and the covering map q is continuous, $X = q(E)$ is compact as the continuous image of a compact space is compact.

Let $x \in X$ be arbitrary. Then there exists an evenly covered open neighborhood U_x of x such that $q^{-1}(U_x) = \bigsqcup_{\alpha \in A} V_\alpha$, where each V_α is open in E and contains a unique point $x_\alpha \in q^{-1}(x)$, which means that x_α is not in any other V_β for $\beta \neq \alpha$. So $q^{-1}(x)$ is a discrete subset of E .

Now we prove that a closed discrete subset of a compact space is finite. Since $\{x\}$ is closed in X and q is continuous, $q^{-1}(x)$ is closed in E . Since E is compact, the subset $q^{-1}(x)$ is compact (every closed subset of a compact space is compact). Since $q^{-1}(x)$ is discrete, for each $x_\alpha \in q^{-1}(x)$, there exists an open set V'_α in $q^{-1}(U_x)$ containing only x_α , so that $\{V'_\alpha\}_{\alpha \in A}$ forms an open cover of $q^{-1}(x)$. Since $q^{-1}(x)$ is compact, there exists a finite subcover $\{V'_{\alpha_1}, V'_{\alpha_2}, \dots, V'_{\alpha_n}\}$ of $q^{-1}(x)$. As there is only one point in each V'_{α_i} , we can infer that $q^{-1}(x)$ is finite. Hence, q is finite-sheeted. □

Problem 7 (11-5). Let M and N be connected manifolds of dimension n , and suppose $q : \widetilde{M} \rightarrow M$ is a k -sheeted covering map. Show that there is a connected sum $M \# N$ that admits a k -sheeted covering by a manifold of the form $\widetilde{M} \# N \# \dots \# N$ (connected sum of $\widetilde{M}$ with k disjoint copies of N). [Hint: choose the ball to be cut out of M to lie inside an evenly covered neighborhood.]

Lemma 1 (Coordinate neighborhoods form a basis for the topology). *The coordinate neighborhoods of a manifold M form a basis for its topology.*

Suppose U is arbitrary open subset of M , now we prove for any $x \in U$, there exists a coordinate neighborhood B satisfying $x \in B \subset U$. By the definition of manifold, there exists a coordinate chart (U_x, φ) of x , where $\varphi : U_x \rightarrow \varphi(U_x) \subset \mathbb{R}^n$ is homeomorphism and $\varphi(U_x)$ is open set in $\mathbb{R}^n$. Since U_x and U are both open set, $V = U \cap U_x$ is open in M which containing x . In this chart, we choose standard closed n -disk $\mathbb{D}^n(\varphi(x), \varepsilon) = \{y \in \mathbb{R}^n \mid \|y - \varphi(x)\| \leq \varepsilon\}$ in $\mathbb{R}^n$, where $\varphi(x) \in \mathbb{R}^n$ and $\mathbb{D}^n \subset \varphi(V)$. We define the closed disk $D_M \subset M$ as the preimage of this closed ball under the coordinate map:

$$D_M := \varphi^{-1}(\mathbb{D}^n(\varphi(x), \varepsilon))$$

By construction, D_M is closed subspace of M homeomorphic to standard closed disk $\mathbb{D}^n(\varphi(x), \varepsilon)$. So its topological boundary ∂D_M is homeomorphic to standard sphere $\mathbb{S}^{n-1}$, and its topological interior $\text{Int}(D_M)$ is homeomorphic to the standard open n -ball $\mathbb{B}^n(\varphi(x), \varepsilon)$. So we define

$$B_M := \varphi^{-1}(\mathbb{B}^n(\varphi(x), \varepsilon)).$$

As the homeomorphic between B_M (where this is $\text{Int}(D_M)$) and $\mathbb{B}^n(\varphi(x), \varepsilon)$, obviously, B_M is open in M , where B_M is a neighborhood of x . Since $\mathbb{B}^n(\varphi(x), \varepsilon) \subset \varphi(V)$, there are $B_M \subset V = U \cap U_x \subset U$ and $x \in B_M$ which means that B_M is coordinate domain containing x . So B_M is coordinate neighborhood of x , coordinate neighborhoods form a basis for its topology.

Proof. Let x is arbitrary point of M , since $q : \widetilde{M} \rightarrow M$ is a k -sheeted covering map, there exists an evenly covered open neighborhood U of x such that $q^{-1}(U) = \bigsqcup_{i=1}^k \widetilde{U}_i$, where each $\widetilde{U}_i$ is open in $\widetilde{M}$ and $q|_{\widetilde{U}_i} : \widetilde{U}_i \rightarrow U$ is a homeomorphism. Also, each $\widetilde{U}_i$ is open in $\widetilde{M}$ and contains a unique point $x_\alpha \in q^{-1}(x)$.

1° **Definition of coordinate neighborhood:** Since Lemma 1 said that coordinate neighborhoods form a basis for the topology and M is connected manifold of dimension n , then for each point $x \in M$, there exists a local coordinate chart (U_x, φ) around x , where $U_x \subset U$ is open neighborhood of x and $\varphi : U_x \rightarrow \mathbb{R}^n$ is homeomorphism. Let

$$\mathbb{D}^n(p, \varepsilon) = \{y \in \mathbb{R}^n \mid \|y - p\| \leq \varepsilon\}$$

denote the standard closed n -ball of radius $\varepsilon > 0$ centered at $p = \varphi(x) \in \mathbb{R}^n$. By choosing ε sufficiently small, we ensure that $\mathbb{D}^n(p, \varepsilon) \subset \varphi(U_x)$. We define the closed disk $D_M \subset U \subset M$ as the preimage of this closed ball under the coordinate map:

$$D_M := \varphi^{-1}(\mathbb{D}^n(p, \varepsilon)).$$

By construction, D_M is a closed subspace of M homeomorphic to the standard closed n -disk $\mathbb{D}^n$. Its topological boundary

$$\partial D_M = \varphi^{-1}(\partial_{\varphi(U_x)} \mathbb{D}^n(p, \varepsilon)) = \varphi^{-1}(\partial_{\mathbb{R}^n} \mathbb{D}^n(p, \varepsilon))$$

where $\partial_{\mathbb{R}^n} \mathbb{D}^n(p, \varepsilon) = \{y \in \mathbb{R}^n \mid \|y - p\| = \varepsilon\} \cong \mathbb{S}^{n-1}$. Obviously, ∂D_M is homeomorphic to the standard sphere $\mathbb{S}^{n-1}$, and its topological interior $\text{Int}(D_M) \subset U$ is homeomorphic to the standard open n -ball $\mathbb{B}^n$. Similarly, choose a closed disk $D_N \subset N$ such that it satisfies $\partial D_N \cong \mathbb{S}^{n-1}$ and $\text{Int}(D_N) \cong \mathbb{B}^n$.

2° Boundary homeomorphism: We define the punctured manifolds with boundaries by removing the topological interiors of the disk:

$$M' = M \setminus \text{Int}(D_M) \quad \text{and} \quad N' = N \setminus \text{Int}(D_N)$$

Since $\partial M' = \text{Cl}(M) \setminus \text{Int}(D_M)$ and $\text{Int}(D_M)$ is open, the $\text{Int}(D_M)$'s complement M' is closed. So the closure of M' is itself which means that $\text{Cl}(M') = M' = M \setminus \text{Int}(D_M)$. By topological complement laws, the interior of a complement is the complement of the closure. Since D_M is closed disk, $\text{Cl}(\text{Int}(D_M)) = D_M$. Thus:

$$\text{Int}(M') = \text{Int}(M \setminus \text{Int}(D_M)) = M \setminus \text{Cl}(\text{Int}(D_M)) = M \setminus D_M$$

Combining these two results yields the boundary of M' :

$$\partial M' = \text{Cl}(M') \setminus \text{Int}(M') = (M \setminus \text{Int}(D_M)) \setminus (M \setminus D_M) = D_M \setminus \text{Int}(D_M) = \partial D_M.$$

Since $\partial D_M \cong \mathbb{S}^{n-1}$ via the coordinate chart, we have $\partial M' \cong \mathbb{S}^{n-1}$. Similarly, we have $\partial N' \cong \mathbb{S}^{n-1}$. (which can infer $\partial M = 0$)

Fix a boundary homeomorphism $h : \partial M' \rightarrow \partial N'$. The connected sum $M \# N$ is defined as the topological adjunction space:

$$M \# N := M' \cup_h N' = (M' \sqcup N') / \sim_h$$

where the equivalence relation is generated by $y \sim_h h(y)$ for all $y \in \partial M'$. ($M \# N$ is defined as a quotient space via the adjunction map, where the equivalence classes partition the disjoint union by identifying each point $y \in \partial M'$ with its image $h(y) \in \partial N'$ under the boundary homeomorphism h .)

3° Definition of the covering space $\widetilde{M} \# N \# \dots \# N$: Since the closed disk D_M is contained entirely within the evenly covered neighborhood U , its preimage $q^{-1}(D_M)$ decomposes into a disjoint union of exactly k closed disks:

$$q^{-1}(D_M) = \widetilde{D}_1 \sqcup \widetilde{D}_2 \cdots \sqcup \widetilde{D}_k,$$

where for each $i = 1, \dots, k$, the restriction $q|_{\widetilde{D}_i} : \widetilde{D}_i \rightarrow D_M$ is a homeomorphism. We define the punctured covering space by removing the interiors of these k disjoint disks:

$$\widetilde{M}' = \widetilde{M} \setminus \bigcup_{i=1}^k \text{Int}(\widetilde{D}_i).$$

The boundary consists of k disjoint components $\partial\widetilde{M}' = \partial\widetilde{D}_1 \sqcup \cdots \sqcup \partial\widetilde{D}_k$. Now, let $N'_1, \dots, N'_K$ be k disjoint topological copies of N' . For each copy, fix a canonical homeomorphism $\phi_i : N'_i \rightarrow N'$. To glue each copy to its corresponding boundary component on $\widetilde{M}'$, we define k boundary homeomorphisms $g_i : \partial\widetilde{D}_i \rightarrow \partial N'_i$ induced by the base gluing map h and the covering map q :

$$g_i := (\phi_i|_{\partial N'_i})^{-1} \circ h \circ (q|_{\partial\widetilde{D}_i})$$

Finally, we construct the manifold obtained by simultaneously gluing $\widetilde{M}'$ to $N'_1, \dots, N'_k$ along the maps $g_1, \dots, g_k$; we denote this manifold by $\widetilde{W}$. Explicitly, $\widetilde{W}$ is defined as the adjunction space

$$\widetilde{W} := \left(\widetilde{M}' \sqcup N'_1 \sqcup \cdots \sqcup N'_k \right) / \{z \sim g_i(z) : z \in \partial\widetilde{D}_i, i = 1, \dots, k\},$$

and we write $\pi : \widetilde{W} \rightarrow M \# N = (M' \sqcup N') / \sim_h$ be the associated map from $\widetilde{W} \cong \widetilde{M} \# N \# \dots \# N$ to $M \# N$.

4° **π is covering map:** To prove π is covering map, we need to prove π is continuous and surjective, and each point $p \in X$ has a neighborhood U that is evenly covered by π .

1. **Well-definedness and continuity:** We define the natural map $\widetilde{\pi}$ from $\widetilde{M}' \sqcup N'_1 \sqcup \cdots \sqcup N'_k$ to $M' \sqcup N'$, which is defined as

$$\widetilde{\pi}|_{\widetilde{M}'} = q|_{\widetilde{M}'} \quad \text{and} \quad \widetilde{\pi}|_{N'_i} = \phi_i \quad \text{for each } i = 1, \dots, k.$$

Let $\pi_{\text{base}} : M' \sqcup N' \rightarrow M \# N$ be the canonical quotient map. To show that the composite map $\pi_{\text{base}} \circ \widetilde{\pi}$ induces a well-defined map $\pi : \widetilde{W} \rightarrow M \# N$ on the quotient space, we check the boundary identifications: for any $z \in \partial\widetilde{D}_i$,

$$\pi_{\text{base}}(\widetilde{\pi}(g_i(z))) = \pi_{\text{base}}(\phi_i(g_i(z))) = \pi_{\text{base}}(h(q(z))) = \pi_{\text{base}}(q(z)) = \pi_{\text{base}}(\widetilde{\pi}(z))$$

Then we have commutative diagram:

$$\begin{array}{ccc} \widetilde{M}' \sqcup N'_1 \sqcup \cdots \sqcup N'_k & \xrightarrow{\pi_{\text{base}} \circ \widetilde{\pi}} & M \# N \\ \downarrow q_{\text{cov}} & \nearrow \pi & \\ \widetilde{W} = (\widetilde{M}' \sqcup N'_1 \sqcup \cdots \sqcup N'_k) / \sim_g & & \end{array}$$

By the characteristic property of the quotient topology(Thm.3.70), $\pi_{\text{base}} \circ \widetilde{\pi}$ induces a well-defined and continuous map $\pi : \widetilde{W} \rightarrow M \# N$.

2. **Surjectivity:** Since the restriction $q|_{\widetilde{M}'} : \widetilde{M}' \rightarrow M'$ and maps $\phi_i : N'_i \rightarrow N'$ are surjective, the composite map $\pi_{\text{base}} \circ \widetilde{\pi}$ is surjective, which directly implies that $\pi : \widetilde{W} \rightarrow M \# N$ is surjective.
3. **Evenly covered by π :** By the definition at the beginning, a neighborhood U of x is evenly covered by q . Let p is arbitrary point of $M \# N$. If $p \in \text{Int}(M')$, then there exists an evenly covered neighborhood U of p in M such that $\pi^{-1}(U_p) = q|_U^{-1}(U_p) = \bigsqcup_{i=1}^k \widetilde{U}_i$, where $U_p = U \cap \text{Int}(M')$ is open in M' and neighborhood U_p is a evenly covered by q . If $p \in \text{Int}(N')$, then since ϕ_i is homeomorphism, there exists an evenly covered neighborhood $U \subset \text{Int}(N')$ of p in N' such that $\pi^{-1}(U) = \bigsqcup_{i=1}^k \phi_i^{-1}(U)$, where each $\phi_i^{-1}(U)$ is open in N'_i and $\pi|_{\phi_i^{-1}(U)} : \phi_i^{-1}(U) \rightarrow U$ is a homeomorphism. If $p \in \partial M' \sim_h \partial N'$, let $p_M \in \partial M'$ and $p_N = h(p_M) \in \partial N'$ be the representatives of p in $M' \sqcup N'$. Since q is covering map, $q^{-1}(p_M) = \{\widetilde{p}_1, \dots, \widetilde{p}_k\} \subset \widetilde{M}'$, where each $\widetilde{p}_i \in \partial\widetilde{D}_i$. Choose an evenly covered neighborhood $V \subset M$ of p_M under q , such that $q^{-1}(V) = \bigsqcup_{i=1}^k \widetilde{V}_i$, where each $\widetilde{V}_i$ is open in $\widetilde{M}'$ and every restriction $q|_{\widetilde{V}_i} : \widetilde{V}_i \rightarrow V$ is a homeomorphism. Let $V_M = V \cap M'$. Since

$h : \partial M' \rightarrow \partial N'$ is a homeomorphism, $h(V_M \cap \partial M')$ is an open neighborhood of p_N in $\partial N'$. Choose an open neighborhood $U_N \subset N'$ of p_N such that $U_N \cap \partial N' = h(V_M \cap \partial M')$. Define $U := \pi_{\text{base}}(U_M \sqcup U_N) \subset M \# N$. Then U is an open neighborhood of p in $M \# N$. The preimage $\pi^{-1}(U) \subset \widetilde{W}$ consists of k disjoint open sets $W_1, \dots, W_k$, where each W_i is formed by gluing $V_i \cap \widetilde{M}'$ and $\phi_i^{-1}(U_N)$ along their matching boundaries via g_i :

$$W_i := \left((V_i \cap \widetilde{M}') \sqcup \phi_i^{-1}(U_N) \right) / \left\{ z \sim g_i(z) \mid z \in \partial \widetilde{D}_i \cap V_i \right\}.$$

Thus, $\pi|_{W_i} : W_i \rightarrow U$ is a homeomorphism for each i , showing that U is evenly covered.

Conclusion: $\pi : \widetilde{W} \rightarrow M \# N$ is a continuous, surjective local homeomorphism, and thus a covering map.

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